



Introduction to Business Database Management

BIS 603/BIS604: BUSINESS DATA MANAGEMENT

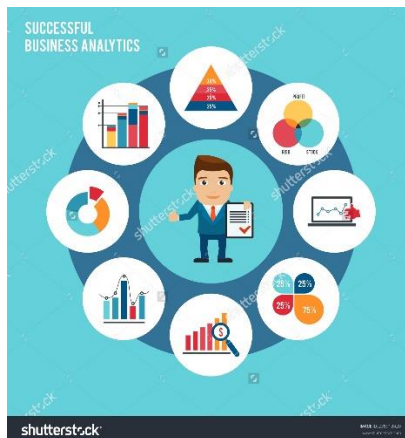
ASSOC. PROF. DR. SUREE FUNILKUL

What is Analysis

- ▶ Analysis is increasingly strategic to businesses and is now central to most business roles.
 - ▶ Every business is an analytics business,
 - ▶ Every business process is an analytics process, and
 - ▶ Every business user is an analysis user.



<https://www.youtube.com/watch?v=IV7mVOI5Gug>



EXPLORING



Data Mining

EXPLAINING



Statistical and
quantitative analysis

EXPERIMENTING



A/B testing
multivariate
testing

FORECASTING



Predictive
analytics

Benefits of Analysis

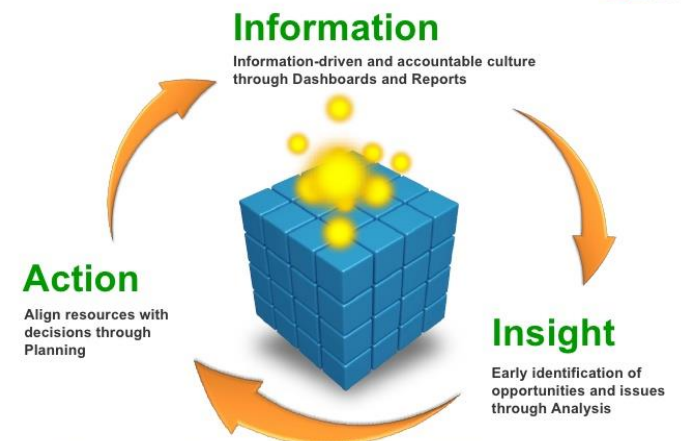
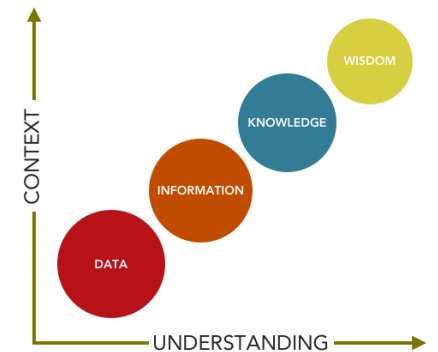
- ▶ Make better business decisions
- ▶ Understand past performance
- ▶ Measure ongoing business performance against KPI
- ▶ Predict future outcome
- ▶ ...
 - ▶ Make customer satisfaction
 - ▶ Know customer needs

Needs

- ▶ Integrate wide and variety of data sources
- ▶ Make fact-based decision from employees, business partners and customers
- ▶ Explain the data by using the analytic tools and techniques

https://www.youtube.com/watch?v=IV1s5NQWN_A

- ▶ Continuing courses: BIS602: Data Analytics, BIS693: Big Data, BIS624: Data Mining



Building Blocks

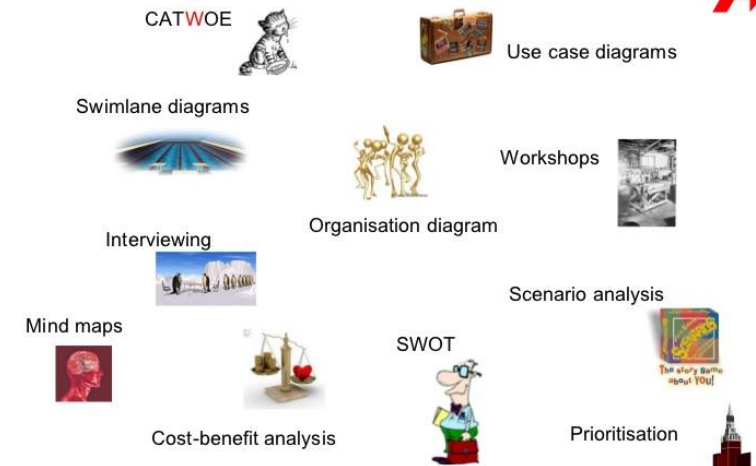
- ▶ Data sources
- ▶ Business needs
- ▶ Data Storage
- ▶ Business analysis tools and techniques

<https://www.fujitsu.com/global/products/computing/storage/>



Foundation for business analytics and optimization. (Modified from: O. Oursin, "IBM business analytics and optimization – The path to breakaway performance," IBM Presentation, November, 2012.)

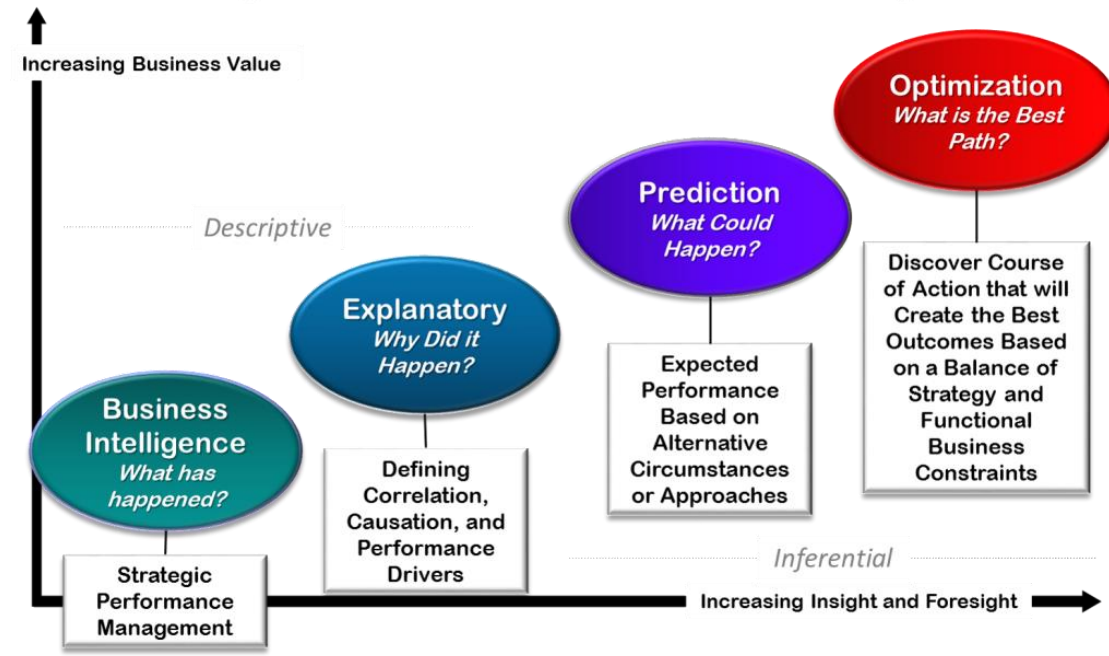
Business Analysis Techniques – The 'first eleven'



© Assist Knowledge Development, 2012

From: Business Analysis Techniques 72 essential Tools for Success – Cadle, Paul and Turner (2010)

Building Blocks of Business Analytics



Design Preview [Prototype]

ORDER FORM

Name

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Phone

E-Mail

Expedition Service

Destination

University Horticulture Society - Membership Enrollment

Personal Information Checkout Confirmation

Personal Information

* Email:

* Forename:

Middle Name:

* Surname:

* Country:

* Address:

* Town/City:

* Post Code:

* Work Telephone:

Mobile Phone:

* Gender:

* Upload Photo:

* Membership Number:

Your Password

Create your own password below (6-20 characters long).

* Password:

* Re-enter Password:

[Contact Info](#) [Terms of Use](#) [Privacy Policy](#)

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Business Form

[HTTPS://WWW.JOTFORM.COM/FORM-TEMPLATES/STUDENTS-REGISTRATION-FORM](https://www.jotform.com/form-templates/students-registration-form)

Database Management

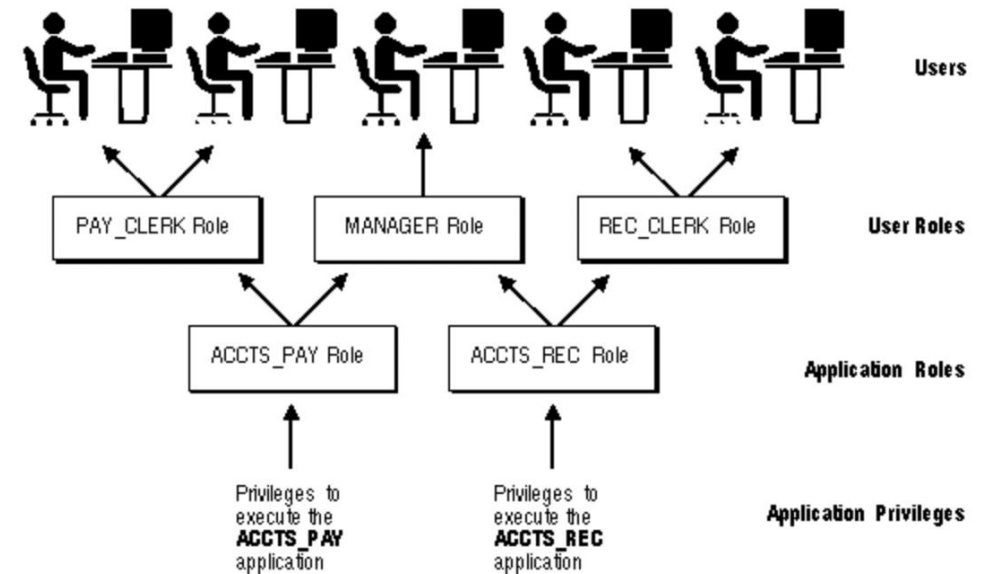
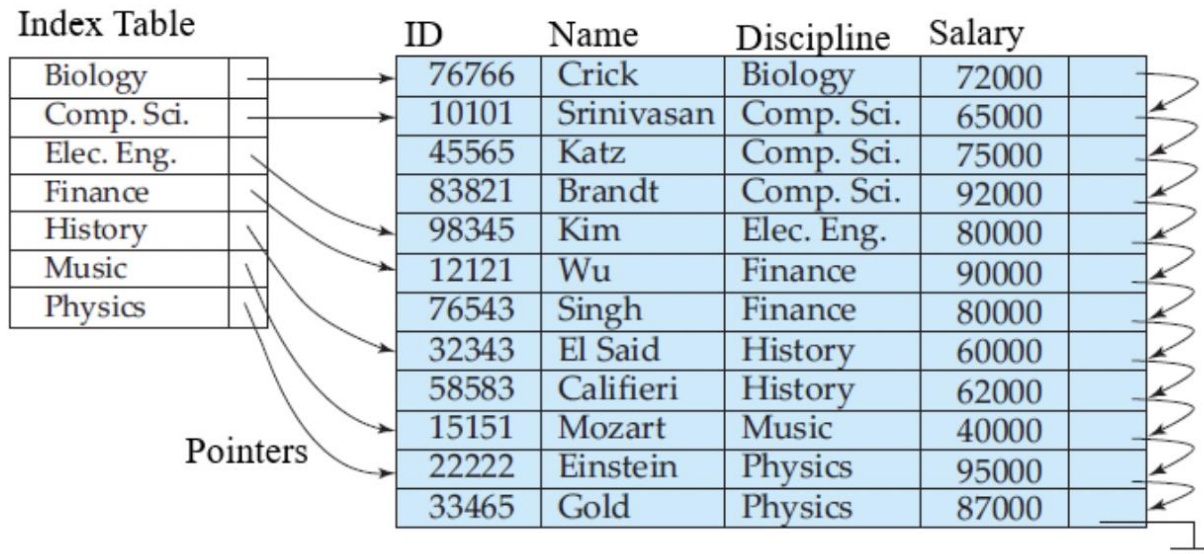
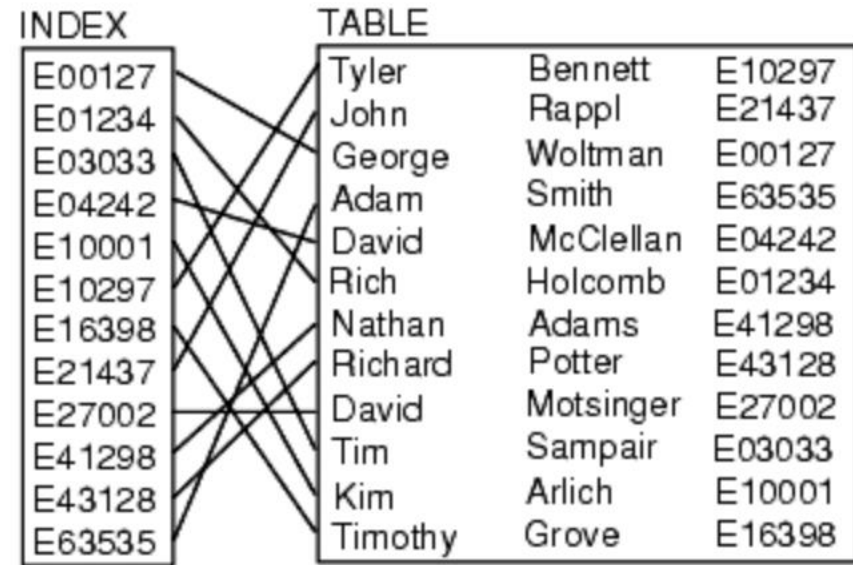
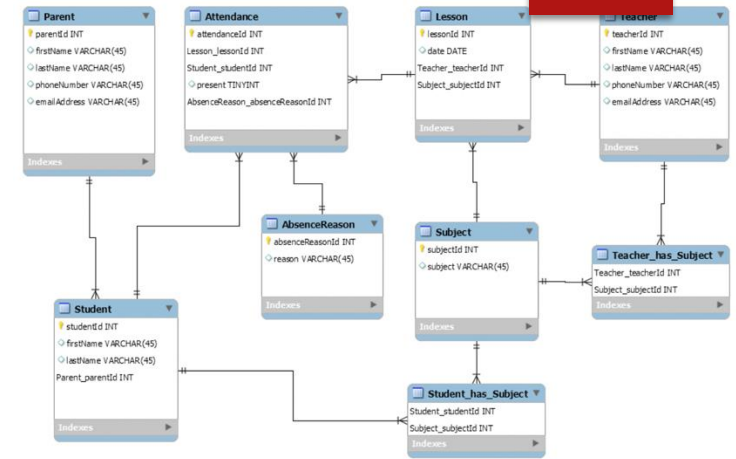
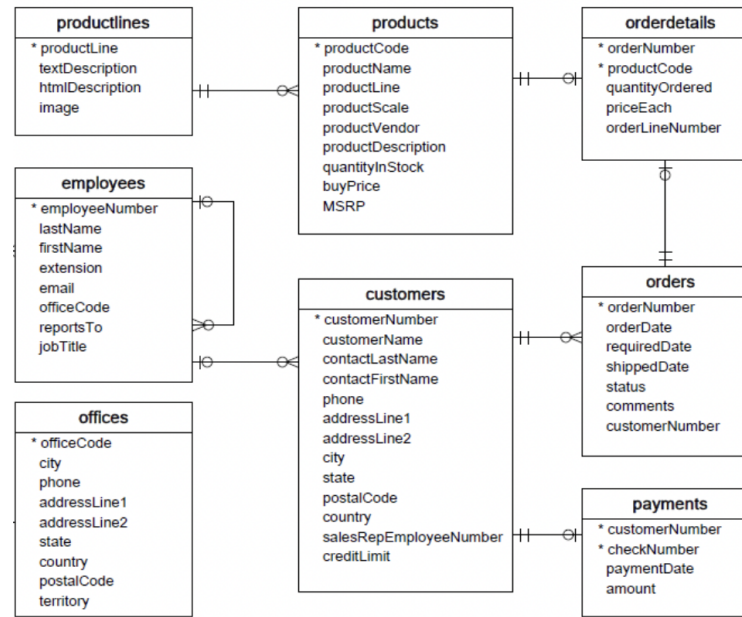
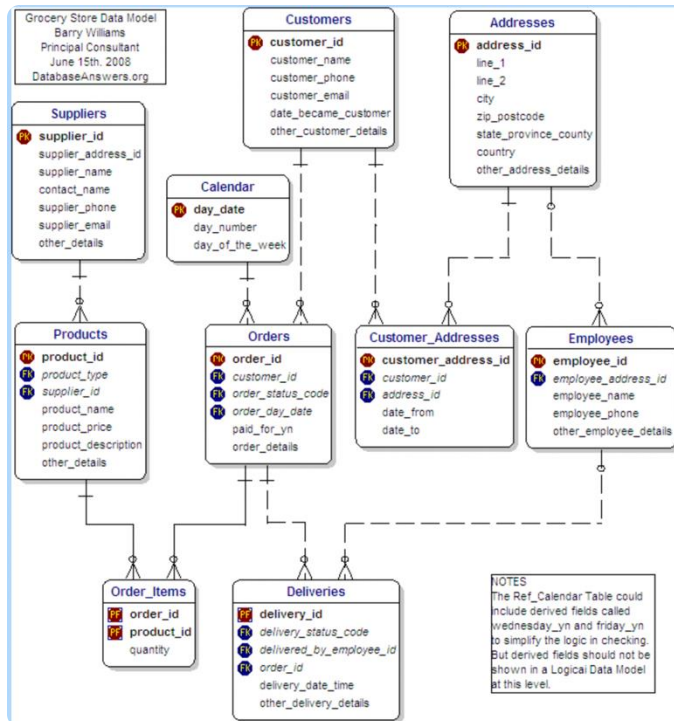
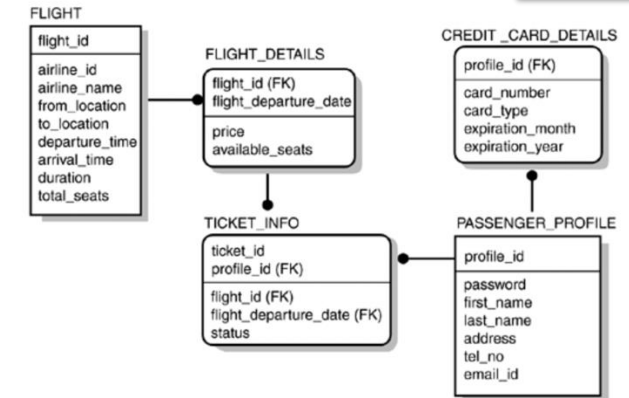
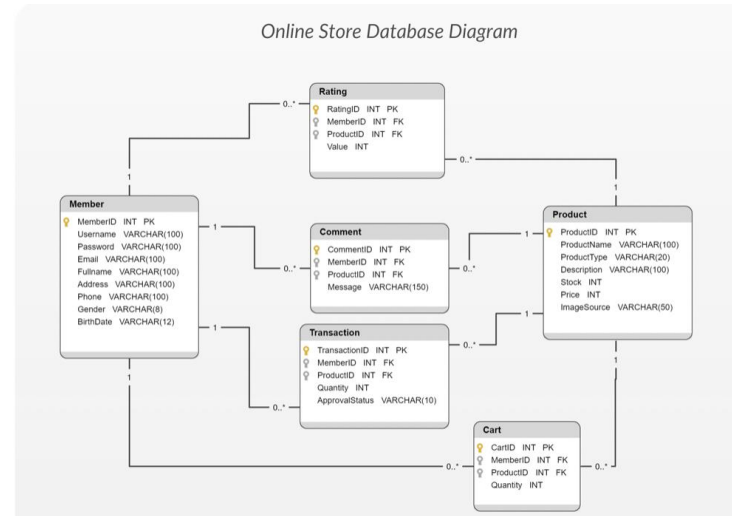
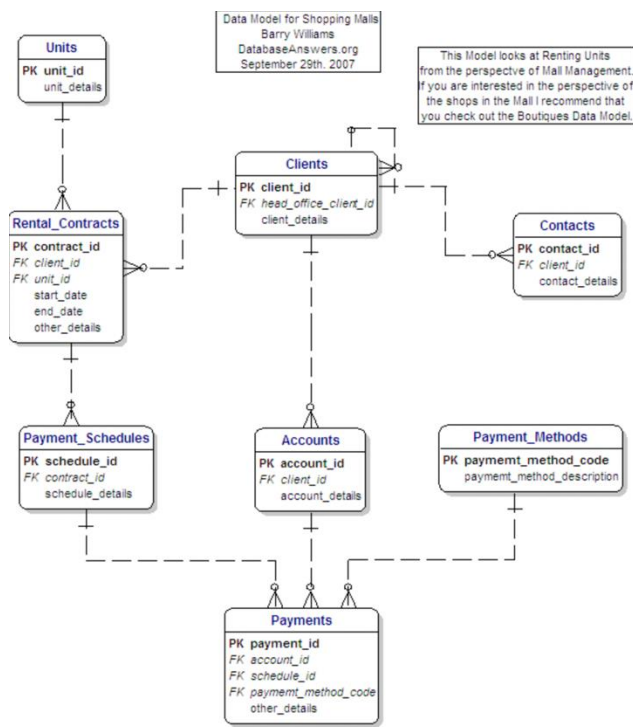


Figure 18 - 1. Common Uses for Roles



Database



Database

Why Database

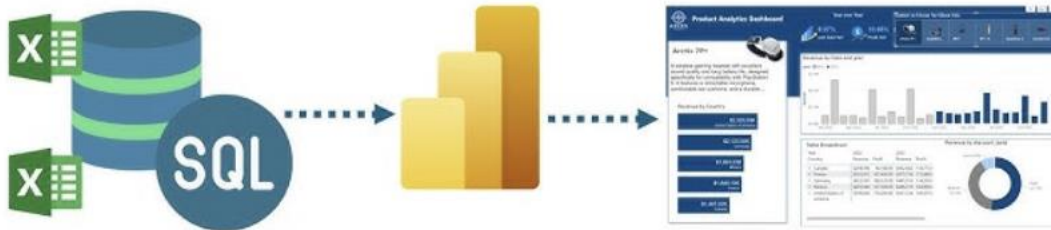
- ▶ Business has some problem
- ▶ Business needs to solve these problems in the best way
- ▶ Business cannot prepare the data for analytics
- ▶ <https://www.youtube.com/watch?v=djEZeF4KTaM>



Data Analytics

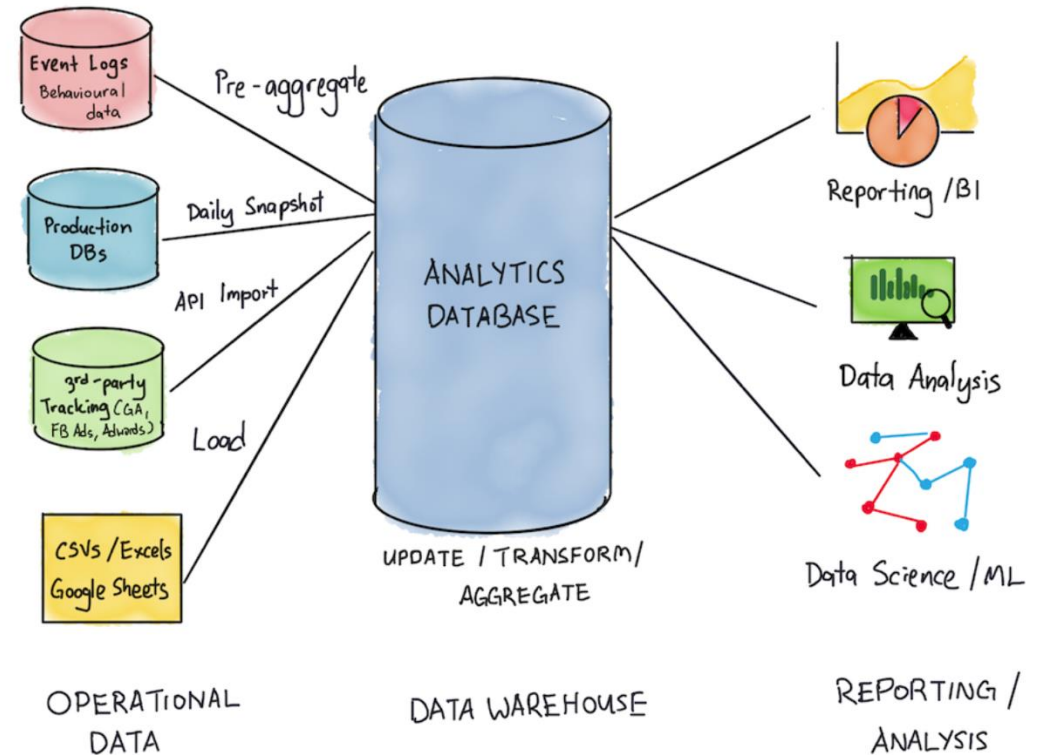
End-to-End Data Analysis

Product Analytics Dashboard



Create a Database → Develop SQL → Build a Dashboard

<https://www.youtube.com/watch?v=6MOyrQLCi3w>



<https://www.holistics.io/books/setup-analytics/a-modern-analytics-stack/>

More information : <https://dataforest.ai/blog/the-importance-of-data-analytics-in-todays-business-world>